Chapter 5

Protecting the Body: The Orbiter's Thermal Protection System

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Developing a thermal protection system (TPS), often called a *heat shield*, was one of the larger challenges facing scientists and engineers as they began seriously looking at lifting-reentry vehicles during the 1950s. In selecting a concept, researchers had to consider the physical, mechanical, chemical, and economic characteristics of the material and the vehicle [1]. Initially, almost all heat-shield research centered on various ablative and radiative metallic materials, and with one exception, it was not until well into the Space Shuttle conceptual development cycle that scientists and engineers seriously considered other materials.

All of the manned orbital space capsules used ablative heat shields, but during the 1950s and early 1960s, most scientists and engineers assumed any advanced TPS would be metallic. One of the larger problems that researchers uncovered was that most metals tended to rapidly oxidize when exposed to high temperatures. The oxidation quickly compromised the material's strength and thermal properties. As a result, researchers expended a great deal of effort trying to find a long-lasting coating that would provide adequate protection for the metal, not affect its mechanical properties, and not have a negative impact on the desired emittance properties. By 1964, researchers concluded there was no "ideal" coating for any given material because each application had different needs. Nevertheless, they had made significant progress toward understanding specific silicide and aluminide coatings for columbium, molybdenum, tantalum, and tungsten. The suborbital Mercury capsules used heat sinks, much like the ICBM warheads of the era [2].

Interestingly, early in the search for a reusable lifting reentry vehicle, a few engineers briefly investigated a ceramic tile concept similar to that ultimately used by the Space Shuttle. During 1951, Lawrence D. "Larry" Bell (1894–1956), Walter R. Dornberger (1895–1980), and Krafft A. Ehricke (1917–1984) were working at the Bell Aircraft Company in Buffalo, New York. One of their projects was a reusable lifting reentry vehicle known variously as BoMi, Project MX-2276, and RoBo; the concept eventually formed the basis for the ill-fated Dyna-Soar program.

In 1954, Bell researchers investigated covering the MX-2276 glider with lightweight bricks, each measuring 10 inches square. The bricks were made from Sil-O-Cel®, first developed by the Celite Products Company (now Celite Corporation) in 1924 and still available in various forms. The bricks had a density of 23.5 pounds per cubic foot, 86 percent of which was silicon dioxide derived from diatomaceous earth. These were generally similar to the firebricks used by Babcock & Wilcox for boilers installed on large ships. The bricks contacted the airframe only along the outer edges, with the beveled center filled with Fiberfrax insulation. A single center attachment consisting of a large countersunk ceramic pin and an Inconel bolt secured each brick. The bricks were spaced from their neighbors with a gap sufficient to accommodate thermal expansion and structural strains. A thin ceramic coating covered the outer surfaces of the bricks for aerodynamic smoothness and waterproofing. It was, essentially, the solution ultimately chosen for the Space Shuttle [3].

It took a while, but NASA Langley finally noticed the ceramic concept and awarded Bell a small contract in August 1965 for continued evaluation. By now, Lockheed had been working on a similar concept for several years under contract to NASA Ames. James N. Krusos led the effort at Bell, which ran through October 1966. Bell researchers selected porous ceramics because of their "advantageous insulation and weight characteristics." The perceived advantages of ceramics included their (1) ability to maintain a smooth and stable outer mold line, (2) need for little or no refurbishment, (3) lightweightness, and (4) relatively low cost, primarily because of their reusability [4].

Oddly, considering that the silicon dioxide Sil-O-Cel fire bricks investigated in 1954 weighed only 23.5 pounds per cubic foot, Bell now estimated that a fused silica ceramic would weigh between 34 and 68 pounds per cubic foot. Combined with support panels and insulation, the TPS weighed 3.28 pounds per square foot of surface area. An active cooling system in the underlying structure weighed another 0.49 pounds per square foot, for a total of 3.77 pounds per square foot. This compared favorably to a contemporary ablative heat shield at 4.0 pounds per square foot [5].

The researchers came away seemingly impressed, thinking ceramics held "promise." They cautioned, however, "that realization of the benefits of highly refractory and stable heat shields requires further refinement of structural analysis and material technologies." The researchers believed ceramics were attractive from a weight standpoint and in regard to component simplicity and refurbishment. The ceramics also maintained a constant outer mold line and vehicle smoothness, unlike ablative heat shields, which changed shape and became rough during entry [6].

The Space Shuttle

Ablators were the departure point for many of the TPS concepts during the Space Shuttle Phase A studies because they were well understood and proven. Researchers studied various materials, including silica/silica-fiber composites; mixed inorganic or organic composites with silica, nylon, or carbon fiber-reinforced resins (phenolics, epoxies, and silicones); and carbon- or graphite-based materials. These ablators had densities ranging from 10 pounds per cubic foot (using microballoons to keep the material lightweight) to 150 pounds per cubic foot (for high-heat protection). Researchers evaluated both charring and noncharring materials, but the need to maintain a precise outer mold line for aerodynamic efficiency usually mandated charring ablators.

One of the most promising ideas to come out of the Phase A studies was to mount the TPS so that it could be easily removed from the vehicle. This permitted technicians to service the systems and structure under the TPS, and allowed the refurbishment of the TPS itself, without taking the entire vehicle out of service for a prolonged period. In this concept, the TPS was mounted on panels that were installed over the basic vehicle structure with an appropriate insulator between. When the vehicle returned from orbit, technicians at the launch site would replace the panels; the old heat shield went to a shop for refurbishment and reuse on a future mission.

Dyna-Soar had eschewed ablators in favor of a metallic heat shield, and this seemed an elegant solution to most engineers. The use of radiative metallic panels covering a cool structural shell with a layer of insulation between would work for all but the hottest regions of the vehicle. Metals are intrinsically durable and therefore capable of extensive reuse, but they have oxidation and strength limitations. A material's ability to resist oxidation largely determines its temperature limitations, and, unfortunately, most of the exotic superalloys tended to oxidize quickly when exposed to high temperatures. At the time, coatings appeared to permit an upper-limit temperature of about 2500°F for 100 cycles, but actual real-world data was lacking [7].

During the follow-on Phase B studies, new materials, including hardened compacted fibers (HCFs) and oxidation-inhibited carbon-carbon, which were neither ablative nor metallic, presented engineers with an entirely different set of options. The most common HCF investigated during these studies was called mullite, although it was not truly mullite, which was a rare chemical found only on the Isle of Mull, off the west coast of Scotland. The glass and steel industries had long used mullite as a refractory material because it exhibited good high-temperature strength, adequate thermal-shock resistance, and excellent thermal stability. Because natural mullite was rare and expensive, scientists developed methods of fabricating synthetic mullite ceramics [8].

Researchers believed that HCF was a promising candidate for space shuttle applications because of its availability and temperature-overshoot capability. The HCF shingles were also lighter than equivalent ablators but somewhat heavier than most metallic concepts. Typical HCF materials were relatively soft, were extremely porous, and had inherently low emittance values. Adhesive bonding had to be used because the shingles would not support mechanical fasteners due to structural weakness. Researchers believed the HCF shingles were potentially useful for 100 flights in the range of 2000 to 3000°F. However, as with the superalloys, researchers needed to develop suitable coatings to protect the HCF before it could be successfully used on any space shuttle. In many ways, the mullite shingles were an indication of things to come [9].

The various studies also investigated a relatively new class of materials, all of which benefited significantly from the investment in Dyna-Soar. Researchers believed that oxidation-inhibited, fiber-reinforced, carbonaceous products offered the potential for a reusable, reasonably cost-efficient, high-temperature-resistant material. Carbon or graphite fibers, usually in the form of cloth, reinforced the carbonaceous material, and the resultant material offered an unsurpassed strength-to-weight ratio at high temperatures. It should be noted that these materials, like the reinforced carbon-carbon actually used on the Space Shuttle, were not insulators and that the back side of the material was essentially as hot as the front side. What the materials provided was a means of creating the appropriate outer mold line on the hottest parts of the vehicle (the nose and the wing leading edge), but some sort of insulating material still needed to protect the primary structure. This insulator was usually a fibrous micro-quartz material characterized by low strength, low density, and low thermal conductivity. Materials typical of this class include Dynaflex®, Micro-Fiber®, and Q-Felt® batt insulation.

REUSABLE SURFACE INSULATION (RSI)

However, researchers were developing another material, although it was similar in many respects to the artificial mullite shingles investigated during Phase B and the conceptual descendant of the Sil-O-Cel firebricks. The

Lockheed Missiles & Space Company in Palo Alto, California, was making progress in the development of a ceramic RSI concept and by December 1960 had applied for a patent for a reusable insulation material made of ceramic fibers. The first potential use came in 1962, when Lockheed manufactured a 32-inch-diameter radome for the Apollo spacecraft. This radome was made from a filament-wound shell and a lightweight layer of internal insulation cast from short silica fibers. However, the Apollo design changed, and the radome never flew. Chemist Robert M. Beasley (1927–1990) led the development of the ceramic tiles ultimately used on the Space Shuttle. Initially, Beasley conceived of the idea while working at the Corning Glass Works in New York, but he did not initiate any real development until he moved to Lockheed during 1960 (Fig. 5.1).

By 1965, researchers at Lockheed had developed LI-1500, the first of what became the space shuttle tiles. This "Lockheed Insulation" (hence the "LI") material had a density of 15 pounds per cubic foot (the "15"), was 89 percent porous, could survive repeated cycles to 2500°F, and appeared to be truly reusable. A sample flew on an Air Force reentry test vehicle during 1968, reaching 2300°F with no apparent problems.

The material was brittle, with a low coefficient of linear thermal expansion, and, therefore, Lockheed could not cover an entire vehicle with a single piece of it. Rather, engineers expected to install the material in the form of tiles, generally 6×6-inch squares. The tiles had small gaps between them to permit relative motion and to accommodate the deformation of the metal structure under them because of thermal effects. A tile could crack as the metal skin under it deformed, so engineers decided to bond the tile to a felt pad and then bond the felt pad to the vehicle. Both bonds used a room temperature vulcanizing (RTV) adhesive. It was a breakthrough, although everybody approached the concept with caution and a certain reluctance. Oddly, nobody seemed to remember the firebricks Bell had proposed in 1954.

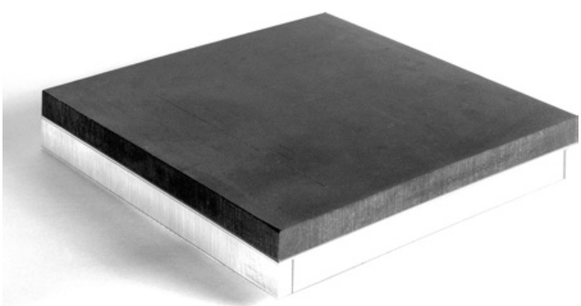

Fig. 5.1 An early LI-15 tile (before it was redesignated LI-1500) that was tested at the MSC in Houston. (NASA)

PHASE C/D DEVELOPMENTS

After long and heated battles within NASA, the DoD, the White House, the OMB, and Congress, President Nixon finally approved the development of the Space Shuttle on January 5, 1972. NASA issued the space shuttle request for proposals (RFPs) on March 17, 1972, to Grumman/Boeing, Lockheed, McDonnell Douglas/Martin Marietta, and North American Rockwell. The RFP required the use of an unspecified reusable TPS, although it allowed ablative material or other special forms of TPSs where beneficial to the program. The orbiter TPS was to be capable of surviving an abort scenario from a 500-mile circular orbit. Unsurprisingly, because the RFP required the contractors to use the NASA-developed MSC-040C concept, the four proposed vehicles looked remarkably similar [10].

Grumman was not completely confident with the Lockheed-developed RSI and proposed adding a layer of ablative material around the crew compartment and the orbital maneuvering system/reaction control system (OMS/RCS) modules during the early development flights to guard against a possible RSI burn-through [11].

Lockheed proposed LI-1500 tiles for the inboard part of the leading edge, using the more expensive carbon-carbon only on the outboard sections. Lockheed believed its TPS could provide 100 normal operational entries at 2500°F or a single-contingency entry at 3000°F [12].

McDonnell Douglas proposed mullite HCF shingles glued to the bottom and side of the orbiter over a thin layer of foam for strain relief. Ablator covered the upper surfaces of the fuselage, wing leading edges, and vertical stabilizer. For some reason, the mullite material was composed of "shingles," whereas the Lockheed RSI used "tiles" [13].

North American was also not completely convinced that the Lockheed-developed RSI would mature quickly enough for the early orbital flights and proposed using HCF shingles manufactured from aluminum silicate, although North American continued to monitor the progress of the silica tiles because they were lighter and more durable. The leading edges and nose cap were reinforced carbon-carbon (RCC) developed by Ling-Temco-Vought (LTV) based on research for the Dyna-Soar program. There was no substantial use of ablator anywhere on the vehicle [14].

ORBITER CONTRACT AWARD

After the stock market closed on July 26, 1972, NASA announced that the Space Transportation Systems Division of North American Rockwell had been awarded the $2.6 billion contract to design and build the Space Shuttle orbiter [15].

Given the progress Lockheed was making developing its RSI, NASA and Rockwell asked the Battelle Memorial Institute to evaluate mullite and

RSI—an evaluation that ultimately favored the Lockheed product. Interestingly, NASA and Rockwell believed that the leeward (top) side of the orbiter would not require any thermal protection. However, in March 1975, researchers from the Air Force Flight Dynamics Laboratory (AFFDL) conducted a briefing for space shuttle engineers on the classified results of the ASSET, PRIME, and BGRV programs. This data indicated leeward-side heating was not particularly severe but easily exceeded the 350°F capability of the aluminum skin. To alleviate this concern, Rockwell proposed to cover areas that never exceeded 650°F, including the upper surfaces of the wings and aft fuselage and payload bay doors, with an ablative elastomeric reusable surface insulation (ERSI) bonded directly to the aluminum skin. Engineers expected, given the relatively benign environment, the ERSI could survive many flights with only minimal refurbishment [16].

Further study showed drawbacks to using the ablator and engineers soon replaced it with a combination of LI-900 tiles and Nomex felt blankets. Black LI-3000 and LI-2200 tiles protected the bottom of the fuselage and wings, the entire vertical stabilizer, and most of the forward fuselage where temperatures exceeded 650°F but were less than 2300°F. The tile thickness varied as necessary to limit the backface temperature to 350°F on the orbiter and 250°F on the OMS pods. As researchers further refined the heating models, the LI-3000 tiles gave way entirely to LI-2200 and finally to a combination of LI-2200 and LI-900 [17].

In the meantime, a problem developed with the tiles themselves. As researchers refined the flight profiles and better understood the aerodynamic environment, engineers began to question whether the tiles could survive the harsh conditions. By mid-1979, it had become obvious that the tiles in certain areas did not have sufficient strength to survive the tensile loads of a single mission. NASA immediately began an extensive search for a solution, which eventually involved outside blue-ribbon panels, various government agencies, academia, and most of the aerospace industry. As Space Shuttle Program Deputy Director LeRoy E. Day recalled, "There is a case [the tile crisis] where not enough engineering work, probably, was done early enough in the program to understand the detail—the mechanical properties—of this strange material that we were using" (Fig. 5.2). It was, potentially, a showstopper [18]. North American manufactured several test articles to evaluate the TPS and its underlying structure; constant damage eventually led NASA to replace the LI-2200 tiles with a small RCC "chin panel."

As engineers spent more time examining the problem, they realized the issue was not with the actual tiles but with how the tiles attached to the underlying structure. Analysis indicated that although each individual component—the tile, the felt strain-isolation pads (SIP) under the tile, and the two layers of adhesives—had satisfactory tensile ability, when combined as a system they lost about 50 percent of their strength. Engineers largely attributed this to

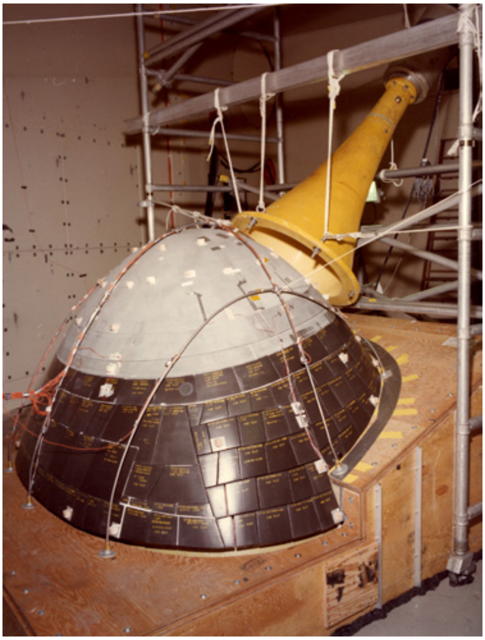

Fig. 5.2 The nose cap and first few feet of the forward fuselage. Note the area on the bottom of the fuselage just behind the nose cap is in the original configuration that used LI-2200 tiles. (NASA)

stiff spots in the SIP (caused by needling) that allowed the system strength to decline to as low as 6 pounds per square inch instead of the baseline 13 pounds per square inch [19].

In October 1979, engineers developed a densification process that involved filling voids between fibers at the inner mold line (next to the SIP pad) with a slurry consisting of Ludox (a colloidal silica made by DuPont) and a mixture of silica and water. The densified layer acted as a plate on the bottom of the tile, eliminating the effect of the local stiff spots in the SIP and bringing the total system strength back up to 13 pounds per square inch (Fig. 5.3) [20]. Early in the program, each vehicle had more than 30,000 individual tiles, although an increased use of blankets reduced this to just 24,000 at the end of the program.

Installing the tiles on the vehicle presented its own issues, and Rockwell quickly ran out of time while *Columbia* was in the manufacturing facility in Palmdale, California. NASA needed to present the appearance of maintaining a schedule, and moving *Columbia* from Palmdale to KSC was a very visible milestone. Rockwell had installed slightly more than 24,000 tiles in Palmdale, and had 6000 to go. Unfortunately, by now it appeared that technicians would have to remove all of the tiles so they could be densified.

The challenge was how best to salvage as many of the installed tiles as possible while ensuring sufficient structural margin for a safe flight. To overcome this almost insurmountable challenge, engineers developed the tile proof test. Technicians induced a stress over the entire footprint of each tile equal to 125 percent of the maximum flight stress experienced at the most critical point on the tile footprint. The proof-test device used a vacuum chuck attached to the tile, a pneumatic cylinder to apply the load, and six pads attached to surrounding tiles to react the load. Because any appreciable tile load might cause internal fibers to break, acoustic sensors placed in contact with the tiles monitored for internal fiber breakage. This testing proved the majority of the tiles were adequately bonded for flight because only 13 percent failed the proof test; those were replaced with densified tiles (Fig. 5.4) [21].

For the next 20 months, technicians at KSC worked 3 shifts per day, 7 days per week, testing, removing, and installing 30,759 tiles on *Columbia*. By the time the tiles were installed, proof tested, often removed and reinstalled, and then re-proof tested, the technicians averaged 1.3 tiles per person per week. In June 1979, Rockwell estimated 10,500 tiles needed to be replaced; by January 1980, technicians had installed more than 9000 tiles, but the number remaining had ballooned to 13,100 because additional tiles failed the proof test or were otherwise damaged. By September 1980, only 4741 tiles remained to be installed, and by Thanksgiving, the number was below 1000. It finally appeared that the end was in sight [22].

This initial debacle caused a great deal of embarrassment for NASA and Rockwell, although it did not seriously delay the first flight because the SSMEs had problems as well. Nevertheless, for the next 30 years, many people believed the tiles were a major maintenance issue for the Space Shuttle program; in fact, they proved to be remarkable durable and trouble free—other than the continual need for pull-tests and waterproofing. The next vehicle will

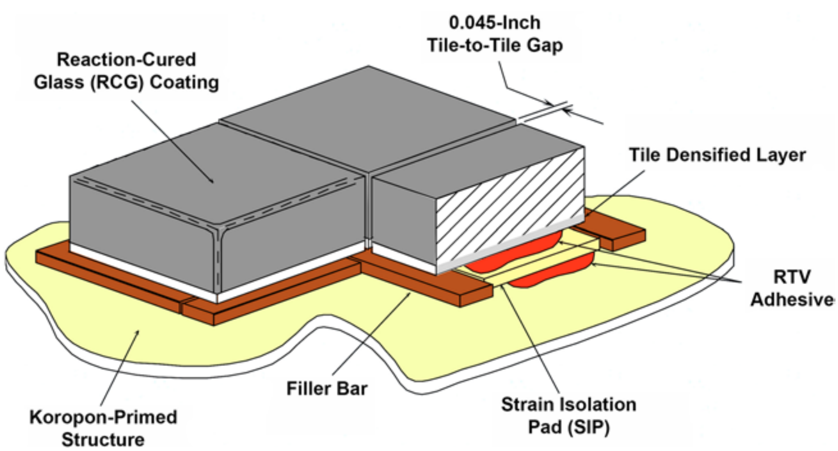

Fig. 5.3 A densified area on the bottom of the tile solved the early problems experienced by the tiles. This illustration shows the major components of the RSI system. (NASA)

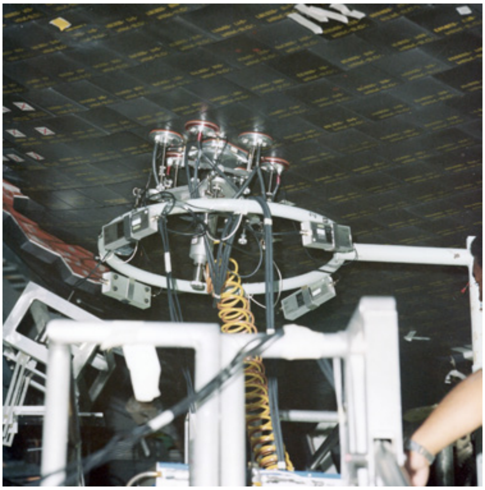

Fig. 5.4 To verify that each tile was well bonded to the orbiter, technicians used this Rube Goldberg device, which used vacuum to gently pull on a tile while reacting the loads to the surrounding tiles. This is *Columbia* on October 3, 1979, in the Orbiter Processing Facility (OPF) at KSC. (NASA)

undoubtedly use something better, but for a first attempt at a reusable TPS, the tiles were a great success.

THE TPS

The TPS protected the aluminum structure of the orbiter during the atmospheric portion of flight on both ascent and entry. During a typical entry, the orbiter endured temperatures in excess of 2300°F and heat loads greater than 66,000 btu per square foot. The orbiter TPS also protected the structure from localized heating from the SSME, SRB, and OMS/RCS exhaust plumes during ascent. The TPS was primarily white on the upper surface and black on the lower surface to control on-orbit heating from solar radiation and to maximize heat rejection during entry [23].

In addition to protecting the structure from heat loads, the TPS outer mold line served as the aerodynamic shape of the vehicle. Rigid control of the step and gap between installed TPS components ensured a smooth surface; excessive steps and/or gaps could cause the boundary layer to transition from laminar to turbulent, resulting in significantly higher heat loads. Even minor steps and/or gaps could result in local overheating, which could slump (that is, melt and deform) tiles or permit subsurface plasma flow that, in turn, could degrade the TPS bond line or underlying structure [24].

The final TPS was composed of RCC, five types of tiles, two types of flexible insulation blankets, thermal barriers, gap fillers, thermal window panes, and thermal seals [25].

RCC

Where temperatures exceeded 2300°F, such as the wing leading edge and nose cap, the orbiter used a composite composed of pyrolyzed carbon fibers in a pyrolyzed carbon matrix with a silicon carbide coating called RCC. In addition, the external tank (ET) forward attach point also used an RCC arrowhead-shaped component because of the pyrotechnic shock environment of the ET separation mechanism. RCC weighed approximately 103 pounds per cubic foot and had an operating range of –250°F to 3000°F. The material had a flexural strength of 9000 psi and a tensile strength of approximately 4500 psi [26].

LTV (later part of Lockheed Martin) of Grand Prairie, Texas, manufactured the RCC, which cost almost $1 million per wing panel. The wing leading edge consisted of 22 RCC panels; because the wing profile changed from inboard to outboard, each RCC panel was unique. Contrary to popular perception, RCC was not an insulator, and the backface was essentially as hot as the frontface. Rather, RCC was simply a material that could withstand the appropriate aero and thermal loads; the underlying structure needed protection from extreme heat by other means [27]. The Leading Edge Structural Subsystem (LESS) was a complex piece of engineering that consisted of RCC panels and the hardware needed to attach them to the wing (Fig. 5.5).

RCC fabrication began with a graphite-saturated rayon cloth impregnated with a phenolic resin that was layed up as a laminate and cured in an autoclave. After curing, the laminate was pyrolyzed to convert the resin to carbon. The laminate was then impregnated with furfural alcohol in a vacuum

Fig. 5.5 RCC panel 8-L (the eighth panel on the left wing) shows the structure that supported the panel as well as the insulation behind it. (NASA)

chamber, cured, and pyrolyzed again to convert the furfural alcohol to carbon. This process was repeated three times until the desired properties were achieved [28].

To provide sufficient oxidation resistance to allow reuse, the outer layers of the RCC were converted to silicon carbide. To accomplish this, technicians dry-packed the RCC with material made of alumina, silicon, and silicon carbide and then placed it into an argon-purged furnace with a stepped-time-temperature cycle to 3200°F. A diffusion reaction occurred between the dry pack and carbon-carbon, and this reaction converted the outer layers to silicon carbide with a whitish-gray color. The piece was then sprayed with tetraethyl orthosilicate and sealed with a glossy overcoat. The RCC laminate was superior to a sandwich design because it was lightweight and rugged; it also promoted internal cross-radiation from the hot stagnation region to cooler areas, thus reducing stagnation temperatures and thermal gradients around the leading edge.

A series of floating joints attached the RCC panels to the wing leading edge to reduce loading caused by wing deflections. A T-seal, also made of RCC, between each panel allowed lateral motion and thermal-expansion differences between the RCC and the orbiter wing.

Because RCC was highly radiative and provided no thermal protection, the adjacent aluminum structure was protected by internal insulation. The forward wing spar and nose bulkhead were protected by a series of RSI tiles installed onto removable carrier panels and Inconel 601–covered Dynaflex batt insulation (called Incoflex) [29].

Five years before *Columbia*, beginning in 1998, engineers started modifying the leading-edge RCC to allow it to withstand larger punctures. The primary concern was impacts from micrometeorites and orbital debris. The original design allowed a 1-inch hole in the upper surface of any panel. But on the lower surface, no penetrations were allowed on panels 5 through 13 because any hole would allow heat from the hot gas flow during entry to quickly erode the 0.004-inch Inconel foil on the Incoflex insulators, exposing the leading-edge attach fittings and wing-front spar to the direct blast of superheated air. The upgrade included additional insulation able to withstand a penetration of up to 0.25 inch in diameter in the lower surface of panels 9 through 12 and up to 1 inch on panels 5 through 8 and 13. To achieve this, technicians wrapped Nextel 440 fabric insulation around the Incoflex insulators—one layer for panels 5 through 7 and 11 through 13, and two fabric layers for panels 8 through 10 (which had the highest-potential heating environment) [30].

Despite the modification, an ascent debris strike on an RCC panel on the left wing during STS-107 resulted in the destruction of *Columbia* during entry. A piece of foam from the bipod ramp on the external tank impacted the wing leading edge. The foam penetrated the panel, resulting in hot gas

Fig. 5.6 A test article used by the CAIB to prove foam could destroy RCC. (CAIB)

impinging on the internal structure during entry. Ultimately, the wing failed and *Columbia* broke up, killing her crew of seven. This led to a reevaluation of the relative strength of RCC, and subsequent testing revealed that it was not as robust as once thought, particularly to direct debris strikes. Nevertheless, it was one of the few materials that could withstand the entry environment and continued to be used until the end of the program, albeit with somewhat more inspection and analysis (Fig. 5.6).

RSI TILES

Tiles were manufactured from one of five substrate materials: LI-900, LI-2200, FRCI-12, AETB-8, or BRI-18. The lightest, LI-900, weighed 9 pounds per cubic foot and made up about 83 percent of the total tiles. Denser LI-2200 tiles weighed 22 pounds per cubic foot and were used on the undersurface of the vehicle, making up 2 percent of the tiles. FRCI-12 weighed 12.5 pounds per cubic foot and made up approximately 13 percent of the tiles, with 8-pound-per-cubic-foot AETB-8 and 18.5-pound-per-cubic-foot BRI-18 making up the remainder [31].

The tile substrate material and coating selection were dependent on the mechanical and thermal requirements of the particular location. For example, tiles located on the upper surface of the forward fuselage (nominally 0.75-inch-thick LI-900) experienced lower temperatures and required less strength than tiles on the nose landing gear door (2- to 3-inch-thick LI-2200, FRCI-12, or BRI-18). The thickness of the tiles varied with heat loads and outer mold line contour requirements from less than 1 inch to more than

5 inches. The substrate material was machined to the desired shape (usually a 6 × 6-inch square) before coating [32].

White LI-900 low-temperature reusable surface insulation (LRSI) tiles were used in selected areas of the fuselage, vertical stabilizer, upper wing, and OMS/RCS pods. These tiles protected areas where temperatures were between 750°F and 1200°F. As originally delivered, *Columbia* and *Challenger* made extensive use of LI-900 tiles on the sides of the fuselage and upper wing surfaces. Later vehicles used blankets in these areas, and *Columbia* was brought up to the same standard during Orbiter Maintenance Down Period (OMDP). *Challenger* was lost before any significant changes could be made.

LRSI tiles were of the same construction as LI-900 tiles and had the same basic characteristics as the high-temperature reusable surface insulation (HRSI) tiles but were cast thinner (0.2 to 1.4 inches). The tiles were produced in 8 × 8-inch squares and had a white reaction-cured glass (RCG) optical and moisture-resistant coating applied 10 millimeters thick to the top and sides. The coating was made of silica compounds with shiny aluminum oxide to obtain optical properties and provided on-orbit thermal control for the orbiter. The LRSI tiles were treated with waterproofing and installed on the orbiter in the same manner as the HRSI tiles. The LRSI tile had a surface emittance of 0.80 and a solar absorptance of 0.32 (Fig. 5.7).

Black HRSI tiles protected areas where temperatures were between 1200°F and 2300°F. The HRSI substrate consisted of low-density, high-purity, 99.8-percent amorphous silica fibers 1 to 2 millimeters thick. A slurry

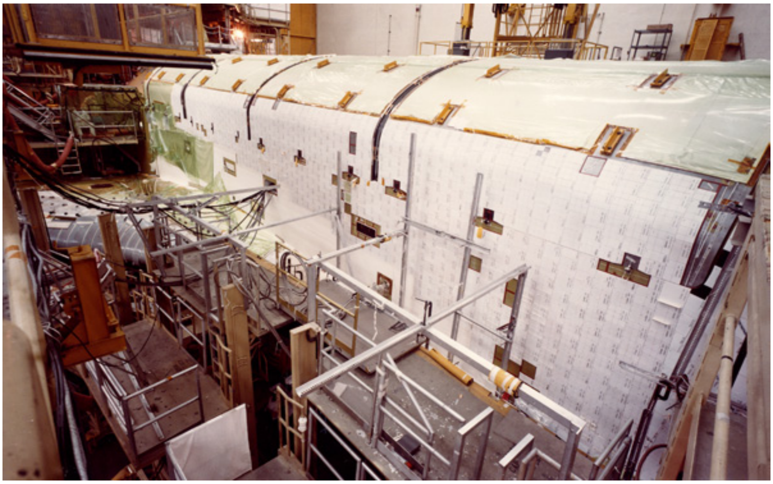

Fig. 5.7 LI-900 tiles used extensively on the sides of the fuselage and upper wing surfaces of Columbia (and Challenger) as delivered while later vehicles used blankets in these areas. Columbia was later refitted.

containing fibers mixed with water was frame-cast to form soft, porous blocks onto which a colloidal silica binder solution was added. Sintering produced a rigid block that was cut into quarters and then machined to the precise dimensions required for individual tiles. HRSI tiles varied in thickness from 1 to 5 inches; generally, the tiles were thicker at the forward areas of the orbiter and thinner toward the aft end. The HRSI tiles withstood on-orbit cold soak conditions, repeated heating and cooling thermal shock, and extreme acoustic environments (163 decibels) at launch [33].

The HRSI tiles were coated on the top and sides with powdered tetrasilicide and sintered borosilicate RCG. The tiles were individually sprayed with 10 to 15 coats of the coating slurry to produce a final coating weight of 0.09 to 0.170 pounds per square foot. The tiles were heated to 2300°F for 90 minutes, resulting in a glossy black coating that had a surface emittance of 0.85 and a solar absorptance of about 0.85. Then, the tiles were waterproofed using heated methyltrimethoxysilane, rendering the fibrous silica material hydrophobic [34].

As an aside, an HRSI tile taken from a 2300°F oven could be immersed in cold water without damage. Also, surface heat dissipated so quickly that an uncoated tile could be held by its edges with an ungloved hand only seconds after removal from the oven while its interior still glowed red.

As LI-2200 tiles were replaced, new fibrous refractory composite insulation (FRCI) tiles were usually used instead. Researchers at NASA Ames developed these later in the program, and they were more durable and resistant to cracking than the earlier HRSI tiles. The tiles were essentially similar to the original HRSI tiles with the addition of a 3M Nextal (AB312—alumina-borosilicate fiber) additive. The resulting composite-fiber refractory material, composed of 20 percent Nextal and 80 percent pure silica, had entirely different physical properties than the original 99.8 percent silica tiles.

The black RCG coating on the FRCI-12 tiles was compressed as it was cured to reduce its sensitivity to cracking during handling and operations. In addition to the improved coating, the FRCI-12 tiles were about 10 percent lighter than the HRSI tiles for a given thermal protection. FRCI-12 tiles demonstrated a tensile strength at least three times greater than that of the HRSI tiles and an allowable temperature approximately 100°F higher.

Late in the program, Boeing Replacement Insulation (BRI-18) began to replace LI-2200 tiles around the landing gear and external tank doors. The Boeing Company in Huntington Beach, California, manufactured the raw substrate from silica mixed with other proprietary elements. The BRI-18 tiles made use of a toughened unipiece fibrous insulation (TUFI) coating developed at NASA Ames. TUFI was the first of a new type of composite known as *functional gradient materials*, in which the density of the material was relatively high on the outer surface and became increasingly lower within the insulation. Unlike the HRSI tiles, in which the tetrasilicide and borosilicate glass coating received little support from the underlying tile, the TUFI outer surface was fully integrated into the insulation, resulting in a more damage-resistant tile; TUFI tends to dent instead of shatter when hit [35].

Because of the constantly changing contours of the orbiter outer mold line, each tile needed to be manufactured to exacting dimensions to fit a specific location on the orbiter. There were two distinctly different types of tile machining, tracing a physical ("splash") model of the cavity on a stylus machine to produce a flight tile, and using a numerically controlled (NC) milling machine to create a tile based on a three-dimensional computer model [36].

Tracing the splash model almost always resulted in a tile that fit the cavity but often resulted in minor outer mold line changes that were not reflected back into the master database. The more precise NC machining was based on data from the master database, which ensured the outer mold line was precisely controlled. Technicians conducted a laser scan of the cavity on the orbiter and fed this data into the master database; oddly, however, this technique often did not produce a tile that fit as precisely as one the manual splash models produced. One of the disadvantages of NC machining was that the initial programming was time consuming, especially if sidewall jogs or nondesign features were required. This time was easily offset for a recurring replacement tile, such as a landing gear door corner tile or a tile adjacent to an RCS thruster that required frequent replacement. Another disadvantage of NC machining was the initial cost of the associated hardware, but this was ultimately offset by the time saved in modeling and the high quality of the finished product [37].

Technicians marked each tile with its part number and a unique serial number using a commercial high-temperature paint called Spearex. The part number was located on the forward or leading edge of the tile to assist in the correct orientation of the tile when it was installed. Because of discontinuities in the thin RCG coating, caused either by processing or flight damage, the tiles required re-waterproofing after each flight. Technicians pneumatically injected each tile with dimethylethoxysilane, which chemically reacted with the silica fibers of the base material and prevented water absorption [38].

Failure to re-waterproof the tiles could have resulted in increased weight from absorbed water or tile damage. The damage would be caused by the absorbed water freezing and subsequently contracting on orbit at cold soak temperatures below –70°F, inducing a fracture at the 1050°F isotherm. During entry, the ice would sublime (change from a solid to a vapor) and complete the failure of the tile at the region previously fractured [39].

Because the tiles could not withstand airframe-load deformation, stress isolation was necessary between the tiles and the orbiter structure. This was provided by SIPs made of Nomex felt. The SIP was bonded to the tiles and then bonded to the orbiter skin using silicon RTV adhesive. However, the SIP introduced stress concentrations at the needled fiber bundles, resulting in localized failure in the tile just above the RTV bond line. To solve this problem, the inner surface of the tile was densified to distribute the load more uniformly. The densification process used a Ludox ammonia-stabilized binder. When mixed with silica slip particles, it became cement, and when mixed with water, it dried to a finished hard surface. A silica-tetraboride coloring agent was mixed with the compound for penetration identification. The densification coating penetrated the tile to a depth of 0.125 inch, and the strength and stiffness of the tile and SIP system were increased by a factor of two [40].

FLEXIBLE INSULATION BLANKETS

White blankets made of coated Nomex felt reusable surface insulation (FRSI) were initially used on the payload bay doors, portions of the upper wing surface, and parts of the OMS/RCS pods. The FRSI blankets protected areas where temperatures were below 700°F. The FRSI blankets nominally consisted of 3 × 4-foot, 0.16- to 0.40-inch-thick sheets, depending on the location. A white-pigmented 92-007 silicon elastomer coating provided aerodynamic-erosion protection and a moisture barrier. The density of the coated material was 5.4 pounds per cubic foot, and it was bonded directly to the orbiter by 0.20-inch-thick RTV silicon adhesive [41].

Nomex felt consisted of noncombustible, heat-resistant aromatic polyamide (aramid) fibers 2 deniers in fineness and 3 inches long. These were loaded into a carding machine that untangled the clumps of fibers and combed them to make a tenuous mass of lengthwise-oriented, relatively parallel fibers called a *web*. The cross-lapped web was fed into a loom, where it was lightly needled into a batt. Generally, two such batts were placed face to face and needled together to form felt. The felt then was subjected to a multineedle process that passed barbed needles through the fiber web in a sewing-like procedure that compacted the transversely oriented fibers into a pad. This needle-punch operation was repeated as many times as required to produce a felt product with specified physical properties. The needled felt was calendered to the proper thickness by passing through heated rollers at selected pressures. The calendered material was heat-set at 500°F to thermally stabilize the felt [42].

After the initial delivery of *Columbia*, NASA developed an advanced flexible reusable surface insulation (AFRSI) as a lightweight replacement for the more fragile LRSI tiles. AFRSI blankets protected areas where temperatures were below 1200°F and aerodynamic loads were minimal, such as the sides of the fuselage and portions of the upper wing surface. AFRSI was a quilted blanket consisting of quartz fibrous batt insulation and woven quartz fiber outer fabric quilted on 1-inch centers to a glass fiber inner fabric. The quilting was accomplished using a Teflon-coated quartz thread with three to four stitches per inch. The sewn quilted fabric blanket was manufactured in 3 × 3-foot squares with an overall thickness of 0.25 to 2 inches. The material

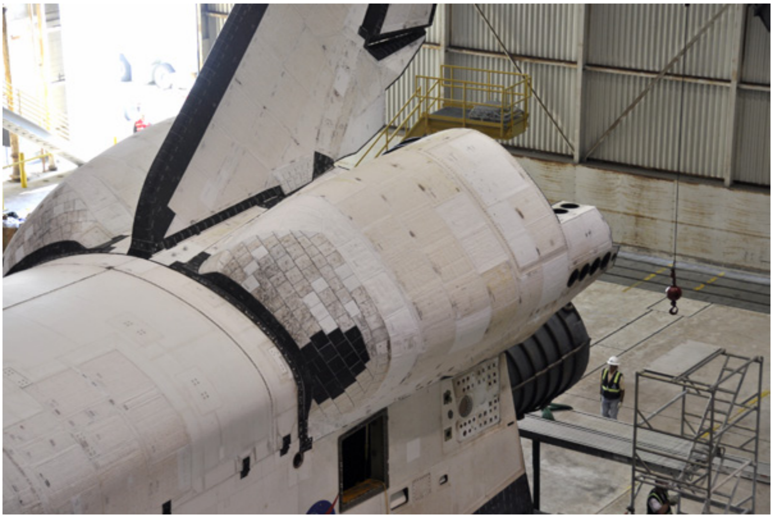

Fig. 5.8 AFRSI blankets. (Dennis R. Jenkins)

had a density of 8 to 9 pounds per cubic foot and was used where surface temperatures ranged from 750°F to 1200°F. The blankets were cut to the planform shape required and bonded directly to the orbiter by 0.20-inch-thick RTV silicon adhesive (Fig. 5.8) [43].

The first application of AFRSI was on the OMS pods of *Columbia* before STS-6. Originally, the OMS pods on *Columbia* and *Challenger* were covered with LI-900 tiles. AFRSI blankets replaced most of these tiles, although the aerodynamic environment on the front edge of the pod dictated that some tiles be retained. The LI-2200 tiles were added in response to a hot vortex that came off the leading edge of the double-delta portion of the wing. AFRSI blankets were subsequently used on *Discovery* and *Atlantis* to replace the majority of the LRSI tiles. The LRSI tiles on the mid-fuselage, payload bay doors, and vertical stabilizer of *Columbia* were replaced with AFRSI blankets during the *Challenger* stand-down. *Endeavour* was delivered with an even greater use of AFRSI, and the other orbiters migrated toward this configuration during normal maintenance as well as major overhauls. At this point, these blankets were used on the fuselage, upper elevons, upper wings, vertical stabilizer, and forward canopy surfaces of each orbiter. However, in an effort to save as much weight as possible for missions to the ISS, much of the AFRSI on the mid-fuselage and aft fuselage, payload bay doors, and upper wing surfaces of the three ISS orbiters was subsequently replaced by the lighter FRSI, saving up to 1400 pounds [44].

As of the last flight of *Atlantis*, in 2011, it (and its two OMS pods) was protected with 20,018 HRSI tiles, 3103 FRCI-12 tiles, 497 LRSI tiles, 325 AETB-18 tiles, 234 BRI-18 tiles, and 1472 blankets of various types. (These numbers varied slightly for each vehicle, with *Columbia* having additional HRSI-22 tiles on the vertical stabilizer [to protect the Shuttle Infrared Leeside Temperature Sensor (SILTS) pod] and additional HRSI-9 tiles on the upper wing leading-edge chines, and *Endeavour* having fewer tiles because of the extensive use of AFRSI.) Of these 24,177 tiles, 19,947 of them (83 percent) remained from the original build. Typically, approximately 20 tiles were damaged on each flight, and the average post-flight refurbishment consisted of about 200 tiles and 10 blankets. However, many of those were replaced for reasons unrelated to their failure, including needing to remove a tile (or blanket) to gain access to the structure or component it covered for maintenance, or replacing the tile (or blanket) with an improved version.

OTHER MATERIALS

Thermal barriers prevented superheated air from flowing through the various penetrations in the orbiter, such as the landing-gear doors and crew hatch. The thermal barriers were tubular Inconel 750 wire mesh filled with Saffil alumina silica fibrous insulation wrapped into a ceramic sleeving with a Nextel AB312 alumina-borosilicate ceramic fabric outer cover. A black ceramic emittance coating was applied to the outer surface, and an RTV silicone adhesive attached the barrier to the structure. Thermal barriers were designed for service in areas seeing temperatures up to 2000°F (Fig. 5.9) [45]. Although most thermal barriers were used around penetrations such as the landing gear doors and crew hatch, similar material was used where the leading-edge RCC met the area protected with LI-2200 tiles.

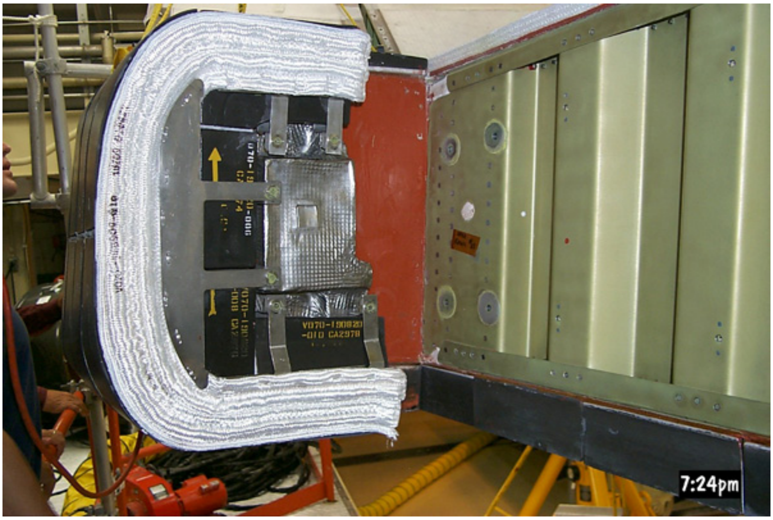

Fig. 5.9 Where RCC panel 22 met the wingtip. (NASA)

Because the tiles were installed with a space between them, it was sometimes necessary to use gap fillers to ensure that heat did not seep between the tiles and impact the aluminum skin. Two types of gap fillers were used: the pillow/pad type and the Ames type. The pillow/pad gap filler was made of a ceramic fabric with alumina silica insulation with two layers of 0.001-inch-thick Inconel foil inside for support. The pad gap filler was stitched with quartz thread, and a black ceramic emittance coating was applied to the outer mold line. A coating of red RTV-560 was applied to the bottom surface as a stiffener to aid with installation into the tile-to-tile gaps. Approximately 5000 pad-type gap fillers, each 0.20-inch thick, were used on the lower fuselage, lower body flap, lower elevons, and vertical stabilizer. The Ames-type gap filler was made from Nextel ceramic fabric that was originally impregnated with a black silicone coating, although this was later replaced with a more durable ceramic coating [46].

Two types of internal insulation blankets were used: fibrous bulk and multilayer. The bulk blankets were fibrous materials with a density of 2 pounds per cubic foot and a sewn cover of reinforced-Kapton acrylic film. The cover material had 13,500 holes per square foot for venting. Acrylic-film tape was used for cutouts, patching, and reinforcements. Tufts throughout the blankets minimized billowing during venting. The multilayer blankets were constructed of alternate layers of perforated-Kapton acrylic-film reflectors and Dacron-net separators. There were 16 reflector layers in all, with the two-cover halves counting as two layers. The covers, tufting, and acrylic-film tape were similar to those used for the bulk blankets [47].

Flags and other markings were painted on the orbiter with a Dow Corning-3140 silicone elastomer, colored with pigments. It was basically the same paint used to paint automobile engines and would break down in temperature ranges between 800°F and 1000°F. Because of this, almost all markings were painted in relatively low-temperature areas of the orbiter.

CONCLUSIONS AND LESSONS LEARNED

Contrary to popular perceptions, the orbiter TPS proved to be quite robust and largely trouble free once the vehicle entered what passed for "operational" service. However, even before the final space shuttle design had been settled upon, some engineers worried about possible damage to the orbiter from debris separating from the booster (in the original two-stage concepts) or external tank (in the design actually built). This was less of a concern in most of the mainstream two-stage designs because the orbiter was mounted well forward on the fly-back booster, minimizing the chances that debris from the booster could impact anything critical (nose cap or wing leading edges) on the orbiter.

This became a greater issue in the final design with the decision to mount the orbiter well back from the nose of the external tank. Initially, the concern was that ice coming off the liquid oxygen tank could impact the orbiter, mostly because, as originally proposed in 1973, the ET did not use any insulation over much of its exterior. However, later that year, engineers decided the orbiter tiles were vulnerable to impact damage, and NASA banned ice and debris shedding from the ET. As early as 1979, NASA commissioned studies into the effects of foam debris on the orbiter, with one stating, "Failure of the HRSI coating was observed to be strongly dependent on the density and size of the projectile" [48].

The spray-on foam insulation used by the ET was mostly successful in eliminating ice debris, although the program accepted that small accumulations would be shed from certain brackets and other areas that could not realistically be insulated. But the solution turned into a problem when pieces of foam began falling off the ET. Eventually, of course, ET foam debris caused the loss of *Columbia* and her crew. The CAIB report offers a detailed discussion of foam, foam debris, and the normalization of deviance that surrounded the loss of *Columbia*.

Even after the loss of the orbiter, many engineers and managers refused to fully accept that something as light as foam could seriously damage the wing leading edge reinforced carbon-carbon, so much of the early investigation centered on possible tile damage, especially near the main landing gear doors.

Fig. 5.10 Even before the first flight, NASA investigated providing a method to repair the thermal protection system while on orbit. At the time, the concern was limited to repairing the HRSI tiles, and during 1980 researchers developed a derivative of MA-25S ablator (used on the X-15A-2) that could be applied to broken to missing tiles by astronauts during an EVA. (NASA)

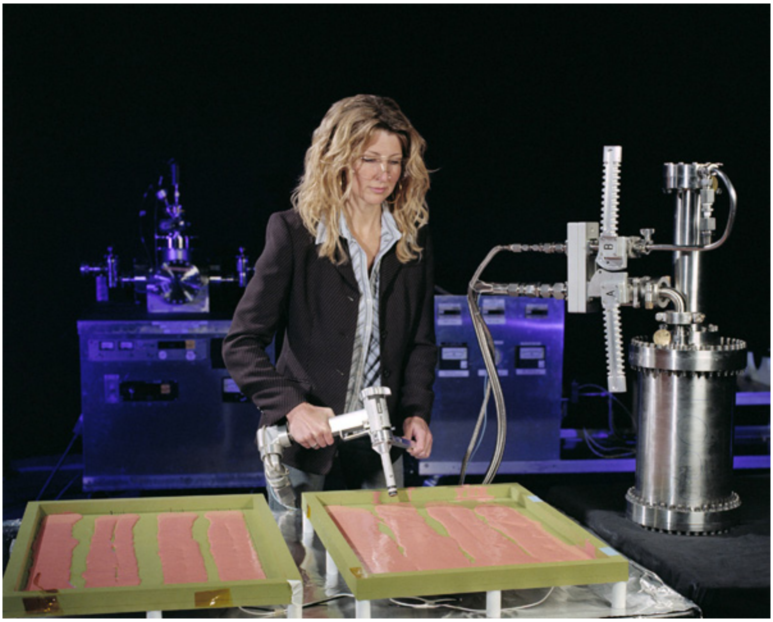

Fig. 5.11 After the Columbia accident in 2004, NASA again looked at providing an on-orbit repair capability. The final system again used an ablator that looked much like the system originally developed in 1980. This time a capability was also developed to repair RCC, although it is unlikely it could have repaired a hole as large as the one that doomed Columbia. (NASA)

Eventually, of course, the laws of physics proved that a comparatively light piece of debris moving quickly could penetrate what seemed like a robust piece of RCC.

The return-to-flight brought many changes, including eliminating much of the questionable insulation on parts of the ET (bipod and PAL ramps) and the development of on-orbit TPS repair techniques (something that had been thought of before STS-1 but abandoned as impractical before the first flight). Even before the first flight, NASA investigated providing a method to repair the TPS while on orbit. At the time, the concern was limited to repairing the HRSI tiles, and during 1980, researchers developed a derivative of MA-25S ablator (used on the X-15A-2) that could be applied to broken or missing tiles by astronauts during an EVA. After the *Columbia* accident in 2004, NASA again looked at providing an on-orbit repair capability. The final system again used an ablator that looked much like the system originally developed in 1980. This time, a capability was also developed to repair RCC, although it is unlikely it could have repaired a hole as large as the one that doomed *Columbia*.

NASA rejected proposals for "robust RCC" and more damage-resistant tiles largely because President George W. Bush had announced in January

2004 that the Space Shuttle would be retired as soon as the assembly of the ISS was complete (Fig. 5.10 and Fig. 5.11) [49].

The orbiter TPS was the first reusable heat shield, and although it failed on STS-107, it proved to be a remarkable piece of engineering. On several occasions, it experienced significant damage yet provided sufficient protection for the orbiter to return home safely. One of these, on STS-27R, came when ablative insulating material from the right-hand SRB impacted *Atlantis* about 85 seconds into the ascent. During the post-landing inspection, NASA found 1 missing tile and more than 700 damaged tiles. Fortunately, the missing tile was located over a dense aluminum mounting plate for the L-band antenna, perhaps preventing a burn-through [50].

Although both tiles and RCC have found use on the Boeing X-37B and Lockheed Martin Orion, most researchers and engineers still appear to be fascinated with the concept of a metallic TPS for any future reusable vehicle.

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